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Data Acquisition

Project: Lambousa Fishing Vessel

Project Reference Number in Metadata:

Date of Digitization: 03/05/2024

Officer(s) Name: Name of responsible officer

Officer Id number: Id number

Officer Profession: profession

Officer Position: position

Officer department: officer dept

Officer telephone: officer telephone

Officer weblink: officer web link

Officer email: officer email

Owner: Limassol Municipality

Stakeholder: Limassol Municipality

Contractor for Digitization - Institution/Organization: UNESCO Chair on Digital Cultural Heritage

Specifications for Data Acquisition & Data Pre-Processing of the: The digital documentation of the boat was carried out based on the cooperation

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agreements between the Municipality of Limassol and the UNESCO Chair within the EU-funded projects H2020 ERA Chair Mnemosyne (https://erachair-dch.eu/) and EU Digital Europe EUreka3D project (https://eureka3d.eu/)

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Reference Image(s) of the object:

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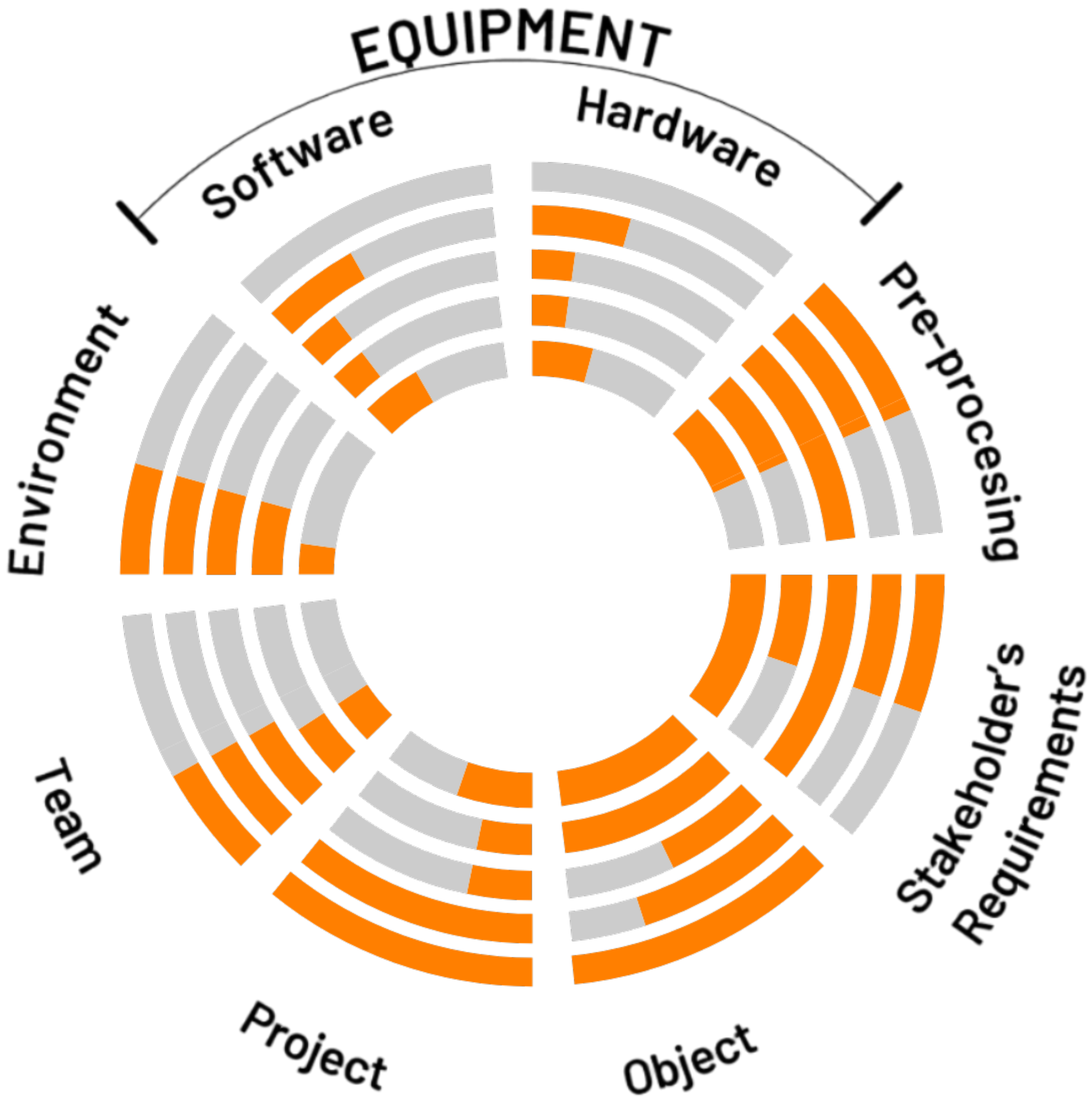
Complexity Chart based on EU VIGIE 2020/654 Study



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Stakeholder's Requirements:

Parameter	Complexity Value	Layer Information	Description
Time (Start&End)	50%	2/5/2022 - 31/10/2023	
Budget (entire project)	50%	0 EUR - Euro	
Purpose of 3D Digitisation	100%	Detailed level	
Location	50%	Address:M28J+75F, Aktaia, Limassol 3013, Cyprus Area: null Village: null City: Limassol District: null Country: Cyprus	
Type	100%	Terrestrial	

Object:

Parameter	Complexity Value	Layer Information	Description
State of Conditions	100%	In decay	
Materials	70%	Wood	Pine Timber, Oak Timber, Metal
Location	50%	34.6655752,33.0295 Area: null Village: null City: Limassol District: null Country: Cyprus	
Dimension & Weight, Touchable, Permissions	100%	Object Dimensions: 25×6.60m Object Weight:	

		Touchable: Yes	
Remedy Options	100%		

Project:

Parameter	Complexity Value	Layer Information	Description
Coordinator/ Management/ Team/ HR Availability	100%	Coordinator Details: Name: Limassol Municipality Surname: Profession: Work ID	
IPR/Deliverables Data	100%	Accuracy: 1cm deviation Transfer: Safety: Security: Long Term Preservation:	
Infrastructure, Transportation	30%	Infrastructure: Transportation: Car	
Accessibility	30%	Obstacles: Dirty road towards shipyard Hazards: Ethical Issues (Permissions): Approachability:	
Budget (co-financing)	50%	0 EUR - Euro	

Data Formats of the Digitized Object:

OBJ - Data Format Details:
Name: Wavefront file
Acronym: OBJ
Extensions: .obj

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Description: OBJ (or .OBJ) is a geometry definition file format first developed by Wavefront Technologies for its Advanced Visualizer animation package. The file format is open and has been adopted by other 3D graphics application vendors. The OBJ file format is a simple data-format that represents 3D geometry alone - namely, the position of each vertex, the UV position of each texture coordinate vertex, vertex normals, and the faces that make each polygon defined as a list of vertices, and texture vertices. Vertices are stored in a counter-clockwise order by default, making an explicit declaration of face normals unnecessary. OBJ coordinates have no units, but OBJ files can contain scale information in a human readable comment line. Object files can be in ASCII format (.obj) or binary format (.mod). This appendix describes the ASCII format for object files. These files must have the extension .obj.
Standard: false
Standard Code:
Link to standard: https://bit.ly/3Dg4Z7r

E57 - Data Format Details:
Name: LIDAR Point Cloud Data File
Acronym: E57
Extensions: .e57
Description: D image file created in the ASTM E57 format, a format used for storing data captured by 3D imaging systems; saves LIDAR (light detection and ranging) data, which was captured by 3D range cameras; enables remote sensing data to be saved in a vendor-neutral format. E57 files can be used for rendering images of real-world objects, such as buildings, atmospheric entities (e.g., clouds), and geological surfaces. This can be useful in construction, surveying, engineering, and research.
Standard: true
Standard Code: ASTM E2807 - 11(2019)
Link to standard: https://bit.ly/3jdAaIz

STL - Data Format Details:
Name: Stereolithography CAD
Acronym: STL
Extensions: .stl
Description: STL is a file format native to the stereolithography CAD software created by 3D Systems company.STL has several backronyms such as "Standard Triangle Language" and "Standard Tessellation Language". This file format is supported by many other software packages; it is widely used for rapid prototyping, 3D printing and computer-aided manufacturing. STL files describe only the surface geometry of a three-dimensional object without any representation of color, texture or other common CAD model attributes. The STL format specifies both ASCII and binary representations. Binary files are more

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common, since they are more compact.
Standard: true
Standard Code: ISO/ASTM 52915:2013
Link to standard: https://bit.ly/3mvQqqp

3DM - Data Format Details:
Name: Open NURBS Initiative 3D Model
Acronym: 3DM
Extensions: .3dm
Description: A 3DM file is an open source file format which is used for 3D graphics software. Developed by the openNURBS initiative, developed by Robert McNeel & Associates, and specifications and libraries are released for free usage by the company. 3DM files contain graphics, metadata details, and other formatting attributes such as surface, points, and curve information. Some of the software that will open, convert, or process 3DM files include Rhinoceros, SAP VEViewer, Moment of Inspiration, and Right Hemisphere Deep View.
Standard: false
Standard Code: ISO/IEC 14496-16:2006
Link to standard: https://bit.ly/388WzR4

FBX - Data Format Details:
Name: Autodesk Filmbox
Acronym: FBX
Extensions: .fbx
Description: FBX (Filmbox) is a proprietary file format (.fbx) developed by Kaydara and owned by Autodesk since 2006. It is used to provide interoperability between digital content creation applications. FBX is also part of Autodesk Gameware, a series of video game middleware.
Standard: false
Standard Code:
Link to standard: https://bit.ly/3yj7Poy

IGES - Data Format Details:
Name: Initial Graphics Exchange Specification
Acronym: IGES
Extensions: .txt
Description: The Initial Graphics Exchange Specification (IGES) is a vendor-neutral file format that allows the digital exchange of information among computer-aided design (CAD) systems.
Standard: true

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Standard Code: NBSIR 80-1978
Link to standard: https://bit.ly/3gyuHKG

Team:

Parameter	Complexity Value (Professional)	Complexity Value (Amateur)	Description
Experience	80.0%	0.0%	
User Qualification for Hardware & Software	80%	0%	
Licenses for mission	80%	0%	
Infrastructure	60%	0%	
Transportation for object and/or team to the special location	60%	0%	

Weather Conditions:

Parameter	Complexity Value	Description
Rain/Snow	40%	
Visibility/Wind Speed	40%	
Air Pressure/Humidity	40%	
Temperature	40%	
Air Pollution	20%	

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Environmental Conditions:

Rain and Temperature		
26 October 2023		
Meteorological Station: Cyprus, Limassol, New Port		
Max. Temperature (°C)	Min. Temperature (°C)	Rain (mm)
28.7	17.8	0.0

Pollution Level (μg/m³)				
Pollutant	Low (1)	Moderate (2)	High (3)	Very High (4)
PM_{10}	0 - 50	50 - 100	100 - 200	> 200
$PM_{2.5}$	0 - 25	25 - 50	50 - 100	> 100
O_3	0 - 100	100 - 140	140 - 180	> 180
NO_2	0 - 100	100 - 150	150 - 200	> 200
SO_2	0 - 150	150 - 250	250 - 350	> 350
CO	0 - 7000	7000 - 15000	15000 - 20000	> 20000
C_6H_6	0 - 5	5 - 10	10 - 15	> 15

Air Pollution	
26 October 2023	
Limassol Traffic Station	
Pollutant	**Date: 9/1/23** Time: 8:00
PM10	39.9
PM2.5	18.3
O3	4.4
NO2	80.7
SO2	4.5

9-13 January 2023					
Limassol Traffic Station					
Pollutant	**Date: 9/1/23** Time: 8:00	**Date: 10/1/23** Time: 8:00	**Date: 11/1/23** Time: 8:00	**Date: 12/1/23** Time: 8:00	**Date: 13/1/23** Time: 8:00
PM_{10}	39.9	70	49.4	19.3	19.3
$PM_{2.5}$	18.3	25.7	17.9	7.4	7.4
O_3	4.4	3	13.1	46.1	46.1
NO_2	80.7	85.9	81.6	40.2	40.2
SO_2	4.5	7.6	3.9	1	1

Pollution Level (μg/m³)				
Pollutant	Low (1)	Moderate (2)	High (3)	Very High (4)
PM_{10}	0 - 50	50 - 100	100 - 200	> 200
$PM_{2.5}$	0 - 25	25 - 50	50 - 100	> 100
O_3	0 - 100	100 - 140	140 - 180	> 180
NO_2	0 - 100	100 - 150	150 - 200	> 200
SO_2	0 - 150	150 - 250	250 - 350	> 350
CO	0 - 7000	7000 - 15000	15000 - 20000	> 20000
C_6H_6	0 - 5	5 - 10	10 - 15	> 15

Air Pollution in Cyprus: https://www.airquality.dli.mlsi.gov.cy/

9-13 January 2023			
Meteorological Station: Cyprus, Limassol, New Port			
Day	Max. Temperature (°C)	Min. Temperature (°C)	Rain (mm)
9	17.6	8.1	0.0
10	18.1	7.0	0.0
11	18.3	10.6	17.8
12	16.5	10.4	39.7
13	15.5	7.1	3.8

Meterological Stations in Cyprus:
https://www.moa.gov.cy/moa/dm/dm.nsf/automaticdata_en/automaticdata_en?OpenDocument

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Software:

Parameter	Complexity Value	Description
Licence for Hardware to be used	0%	
Precision of multisensor system under different environment conditions	40%	
Usability - Communication/Transfer of Data/Battery/Available Storage	20%	
Efficiency - Speed of Data Acquisition in relation to Software & Hardware - Accuracy	20%	
Sensor Integration	40%	

Hardware:

Parameter	Complexity Value	Description
Licence for Hardware to be used	0%	
Precision of multisensor system under different environment conditions	40%	
Usability - Communication/Transfer of Data/Battery/Available Storage	20%	
Efficiency - Speed of Data Acquisition in relation to Software & Hardware - Accuracy	20%	
Sensor Integration	40%	

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Pre-processing:

Parameter	Software	Hardware	Description
Layer 1	100.0%	10.0%	null
Layer 2	100.0%	10.0%	null
Layer 3	100.0%	100.0%	null
Layer 4	100.0%	10.0%	null
Layer 5	100.0%	10.0%	null